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Lesson Notes

MODULE 5 | LESSON 1

THE KELLY CRITERION AND THE OPTIMAL GROWTH STRATEGY

Reading Time	120 minutes
Prior Knowledge	Traditional portfolio theory and utility theory.
Keywords	behavioral finance, bounded rationality, satisfice, prospect theory, cognitive biases, emotional biases.

In the last module, the focus was behavioral finance and how we should incorporate what we know about human behavior into our portfolio construction techniques.

In this lesson, we take up again the topic of utility theory from earlier in this course, which is also grounded in psychology like behavioral finance. The insights into people's feelings about winning and losing (or greed and fear, if you will) directly inform the two main portfolio construction techniques we discuss in this module, namely, the Kelly criterion and risk parity.

We'll see how the Kelly criterion is a fairly aggressive strategy in that end-of-horizon higher returns are sought at the cost of some steep losses along the way.

Risk parity, on the other hand, at least in its original form, seeks to avoid large drawdowns and minimize the variance, by paying little if any attention at all to the expected returns.

Both of these strategies, as we will see, can be extended in the other direction: We can cap the leverage implied by the Kelly criterion, just as we can add higher returns to the objective function of the risk parity strategy.

1. Introduction to the Kelly Criterion

The Kelly criterion can be described as a theory of how to optimally bet during favorable gambling games. Note the request that the gamble be favorable. What does it mean that a bet is favorable? It means that if X_n is chosen to denote the player's wealth or capital after n bets, there exists at least a strategy such that $P[\lim_{n \rightarrow \infty} X_n = \infty] > 0$, i.e., there is a positive probability to win an infinite amount of money in the long run.

It is wise to become familiar with the notion of Kelly criterion by using a relatively simple setup and then build up the difficulty level.

1.1 Binomial game. Simple version

Arguably, the easiest setup is that of binomial game such as coin tossing. Let's say that we win if the coin turns up Heads and lose if the coin turns up Tails, instead. Furthermore, if p is the probability of winning, we assume that the coin is biased (i.e. $p > 1/2$). Finally, we assume that we are facing an opponent who is infinitely rich (the market) and willing to wager even money on bets.

The rules of the game are simple: the bets are independent binomial trial of the biased coin. Let $q = 1 - p$ be the probability of losing. Assume that we start with an initial capital of X_0 . We are going to bet an amount B_i on the i -th trial, for $i = 1, 2, \dots$ and X_n represents our wealth after each game for $n = 0, 1, 2, \dots$

Our goal is to determine the optimal amount B_i to bet on the i -th trial. However, we don't want just any strategy and we look for one that produces a situation where

$$P[\lim_{n \rightarrow \infty} X_n = +\infty] > 0,$$

$$P[\lim_{n \rightarrow \infty} X_n = 0] \neq 1,$$

i.e, we want a strategy where we have a positive probability to end up with infinite wealth and we do not end up ruined with probability 1 in the long run.

Strategy 1.

In treating this example we will use the same notation as in Thorp (1969) so that if a student desired to read the full original paper, he or she will not have to get familiar with a new set of notations.

Let $T_k = 1$ if we won in the k -th trial and $T_k = -1$ if we lost our bet in the k -th trial. Also let S, F denote the number of wins and losses on n trials. Clearly $F + S = n$. We are going to attempt to maximize $E[X_n]$ with the understanding that we cannot get ruined with probability 1. That would be a useless game to play, right? If we let $T_k = 1$ if we record a win in the k -th trial and $T_k = -1$ if the k -th trial resulted in a loss, then we can write

$$X_k = X_{k-1} + T_k B_k$$

for $k = 1, 2, \dots$ as one can easily verify. By repeatedly substituting backward in this formula we get also

$$X_n = X_0 + \sum_{k=1}^n T_k B_k.$$

Therefore,

$$E[X_n] = X_0 + \sum_{i=1}^n E[T_i B_i] = X_0 + \sum_{i=1}^n (p - q) E[B_i].$$

This should be pretty obvious, but since T_k and B_k are independent we have

$E[T_k B_k] = E[T_k] E[B_k]$ and $E[T_k] = 1 * p + (-1) * q = p - q$ from which the formula above follows.

In our setup $p - q > 0$, and thus, in order to maximize $E[X_n]$ we would want to maximize $E[B_k]$ at each trial. In other words, to maximize expected gain we should bet all of our resources at each trial. This means that we choose the strategy where $B_1 = X_0, B_2 = 2X_0, \dots$. This would indeed maximize $E[B_k]$ at each trial. However, since the probability of ruin is $Pr[X_n = 0] = 1 - p^n$, this strategy also yields $\lim_{n \rightarrow \infty} 1 - p^n = 1$; i.e. the probability to be eventually ruined is equal to 1.

We do not accept this strategy and our first attempt did not work.

Strategy 2.

This time we play using a strategy that minimizes the probability of being eventually ruined. This would require that we make the smallest bet possible on each trial and that means $B_k = 0$ for each $k = 1, 2, \dots$. Then, the probability of being eventually ruined is now zero, but I bet that nobody likes this betting strategy either as it is akin to not gambling at all and we stand no chance to reach an infinite win in the long run.

Strategy 3.

We just conclude that we must aim somewhere in between these two extreme approaches. To this purpose we introduce a fraction, f , such that $0 < f < 1$ (we remove the end point or we end up with one of the two betting strategies above and which we deemed not satisfactory). Therefore, our bets are determined by $B_k = fX_{k-1}$ and, after n trials, our wealth would be

$$X_n = X_0(1 + f)^S \cdot (1 - f)^F.$$

We need to stop for a moment and make one technical remark. We need to redefine the notion of "ruin". If for $0 < f < 1$, we used $Pr[X_n = 0] = 0$ for every n , then we are guaranteed that we will never get ruined technically speaking. This condition is too strong and we replace the definition of ruin with one more suitable for applications and that serves the same practical purpose.

$$\lim_{n \rightarrow \infty} P[X_n \leq \epsilon] = 1$$

for an arbitrarily small $\epsilon > 0$.

With this in mind, we point out that Kelly noticed that, since

$$e^{n \ln(X_n/X_0)^{1/n}} = \frac{X_n}{X_0}$$

we can define $\ln(X_n/X_0)^{1/n}$ as the exponential rate of increase per trial and furthermore:

$$\ln(X_n/X_0)^{1/n} = \frac{S}{n} \ln(1 + f) + \frac{F}{n} \ln(1 - f).$$

Kelly chose to maximize the expected value of the growth rate coefficient, which we call $G(f)$; i.e.,

$$G(f) := E[\ln(X_n/X_0)^{1/n}] = \frac{E[S]}{n} \ln(1 + f) + \frac{E[F]}{n} \ln(1 - f).$$

Since $E[S] = np$ and $E[F] = nq$, we have

$$G(f) := p \ln(1 + f) + q \ln(1 - f)$$

which we can maximize with respect to f . To this purpose, we notice that

$$G(f) = \frac{1}{n} E[\ln(X_n)] - \frac{1}{n} \ln(X_0)$$

and for n fixed, maximizing $G(f)$ is equivalent to maximizing $E[\ln(X_n)]$ because X_0 is given. At this point our probabilistic problem has been reduced to one of just calculus. The first derivative is:

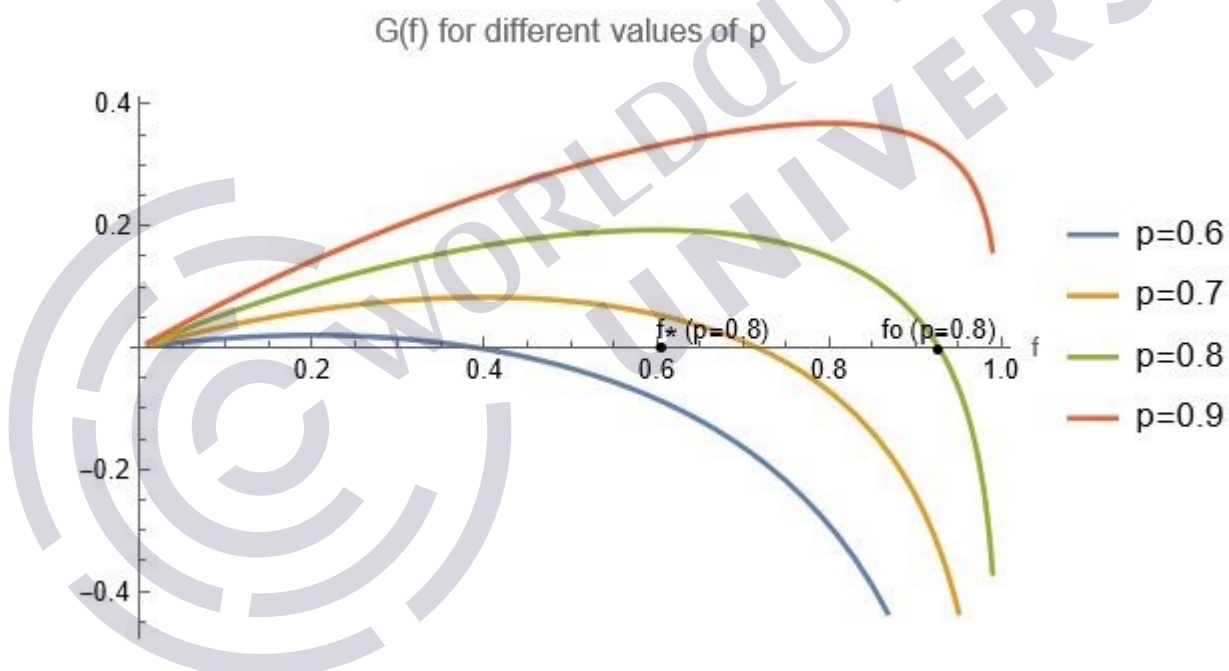
$$G'(f) = \frac{p - q - f}{(1 + f)(1 - f)} = 0$$

which yields that the $f^* := \operatorname{argmax}\{G(f)\} = p - q$. One should verify that $G''(f^*) < 0$ as well for $f \in (0, 1)$. And you can check that this is in fact the case.

This solves part of the problem. We also wanted $\lim_{n \rightarrow \infty} \Pr[X_n = +\infty] = 1$. Well, this is a bit more technical, but there is a result (see for instance E.O. Thorp's paper listed in the reference section), that shows that if $G(f) > 0$, then $\lim_n X_n = +\infty$ almost surely (with probability one). On the other hand, if $G(f) < 0$, then $\lim_{n \rightarrow \infty} \Pr[X_n < \epsilon] = 1$ for each $\epsilon > 0$ arbitrarily small.

In our case the $G(f)$'s looks like in the plot below and so the point f^* achieves even this goal.

Figure 1. Plots of $G(f)$ for different values of the probability to win p and the two relevant points f^* and f^o for $p = 0.8$.



We should notice an interesting feature. Based on the figure above, as the probability of winning gets higher the optimal solution shows that we should bet a larger fraction of our current wealth. That makes sense: the more favorable the game, the more risk we can afford to take. f^o is the point where $G(f) = 0$ and if we bet a fraction $f \geq f^o$, then ruin is unavoidable.

1.2 Binomial game, but monetary wins and losses are not the same

A slight modification of the problem above is to assume that the gamble pays an amount b for each \$1 waged in case of a win and the gambler loses a for each \$1 bet in case of loss.

It is easy to check (just follow the same steps as above) that in this case:

$$G(f) = p \ln(1 + fb) + q \ln(1 - fa).$$

Then

$$G'(f) = \frac{pb}{1+fb} - \frac{qa}{1-fa}.$$

If we solve the equation $G'(f) = 0$ we find (keep in mind that $p + q = 1$):

$$\begin{aligned}(1 - af)bp - aq(1 + fb) &= 0 \\ bp - abfp - aq - abfq &= 0 \\ bp - a(1 - p) - abf(p + q) &= 0 \\ bp - a + ap - abf &= 0 \\ abf &= (a + b)p - a.\end{aligned}$$

Hence

$$f^* = \frac{(a + b)p - a}{ab} = \frac{(a + b)p - a(p + q)}{ab} = \frac{bp - aq}{ab} = \frac{p}{a} - \frac{q}{b}.$$

From the last formula it is easy to see that whenever $a = b = 1$, we get the previous formula, i.e., $f^* = p - q$. In particular, while the previous formula was only a function of p , the new formula for f^* shows that the optimal bet depends again on p , but also on the values of a and b .

1.3 Gambles with many outcomes

Binomial games are relatively easy. Suppose now that the gambles have not just many possible outcomes, but infinitely many. Then we need a continuous distribution of possible outcomes. In real life it is not so, but it is convenient to assume that they can take uncountable many values.

For instance. We purchase a stock for \$100/share and we anticipate that the possible value of the price in one year is distributed as a uniform random variables with possible values ranging from \$20 to \$200, i.e., we could lose 80% or double our investment as extreme outcomes.

It is more convenient to work with outcomes in terms of unit bet, i.e., \$1. Let A be the distribution of our possible returns. Then A has a uniform distribution over the interval $(-0.8, 1)$, i.e., $A \sim U(-0.8, 1)$. Let F be the associated probability distribution. Then $dF(s) = U_A(s)ds$ where $U_A(s)$ is the probability density function for the $U(-0.8, 1)$ random variable and thus $U_A(s) = 10/18$ (Check it as an easy exercise) if $s \in (-0.8, 1)$ and $U_A(s) = 0$ otherwise. We assume that there are no fees, brokers costs, taxes, etc. It would not be a terrible complication, but we are after the general idea.

Now, if we start with an investment of V_0 at time 1 the investment is worth $V_0(1 + fs)$ where f is the fraction bet for each unit. So, using again the definition of $G(f)$ as given above, we get that in this case:

$$G(f) = \int_{-0.8}^1 \ln(1 + fs) \cdot \frac{10}{18} ds.$$

This is not an easy integral. However, we do not need to find the integral and it is enough to calculate $G'(f)$. Using Calculus we get

$$G'(f) = \frac{10}{18} \int_{-0.8}^1 \frac{s}{1+fs} ds = \frac{10}{18f} \int_{-0.8}^1 \frac{fs}{1+fs} ds$$

and, finally, if we use the following algebraic identity $\frac{fs}{1+fs} = 1 - \frac{1}{1+fs}$, we can rewrite the previous formula as

$$G'(f) = \frac{10}{18f} \left[\int_{-0.8}^1 ds - \int_{-0.8}^1 \frac{1}{1+fs} ds \right]$$

At this point the integration is not too complicated.

$$G'(f) = \frac{10}{18f} \left[s \Big|_{-0.8}^1 - \frac{\ln(1+fs)}{f} \Big|_{-0.8}^1 \right].$$

Solving $G'(f) = 0$ we find

$$\frac{18}{10} = \frac{1}{f} \ln \left(\frac{1+f}{1-\frac{8f}{10}} \right).$$

We cannot solve it analytically, but we can solve it numerically and this yields $f^* \approx 0.40$.

Therefore, if we can continue to make similar advantageous bets in the foreseeable future, we should wage 40% of our current capital on each bet.

Numerically we could find $G(0.40)$ and also find the point f^o such that $G(f^o) = 0$. In particular, if we bet a fraction $f > f^o$, then ruin would be certain in the long run. Using some routine or even just some old-fashioned trial and error, we find that if we were to bet a fraction of 0.72 or more, ruin is certain.

In this way we have seen the mathematics related to Kelly criterion both in a simple binomial settings and in a continuous setting (even if we used a simple distribution like the uniform). Different gambles require different setups, but the general idea stays the same.

2. Kelly criterion for log-normal diffusions for stock prices

We are now ready for a situation where we apply the relevant theory to the stock market. More specifically, we are going to solve a problem that involves a log-normal diffusion for stock prices and we will do so by starting with a binomial problem and then we will get the result for the diffusion using a limiting process.

To this purpose, suppose that a random variable X represent some assets returns and that

$$P[X = \mu + \sigma] = \frac{1}{2} = P[X = \mu - \sigma].$$

We also assume that our initial capital is V_0 , r represents the risk-free rate of return, and f is the fraction of wealth allocated to the risky assets. Then,

$$V_1(f) = V_0(1 + (1-f)r + fX) = V_0(1 + r + f(X - r)).$$

Now, we calculate the expected logarithm returns $V_1(f)/V_0$, i.e.,

$$G(f) = E\left[\ln\left(\frac{V_1(f)}{V_0}\right)\right] = \frac{1}{2}\ln(1+r+f(\mu+\sigma-r)) + \frac{1}{2}\ln(1+r+f(\mu-\sigma-r)).$$

Being clear about our basic setup element. we divide the interval of time in n equal independent steps and we keep the mean and the variance the same overall. In practical terms, this simply means that instead of a single return X , we consider now n different returns $X_i, i = 1, 2, \dots, n$ where

$$P\left[X_i = \frac{\mu}{n} + \frac{\sigma}{\sqrt{n}}\right] = \frac{1}{2} = P\left[X_i = \frac{\mu}{n} - \frac{\sigma}{\sqrt{n}}\right].$$

Then

$$\frac{V_n(f)}{V_0} = \prod_{i=1}^n \left(1 + \frac{r}{n} + f\left(X_i - \frac{r}{n}\right)\right),$$

and

$$\ln\left(\frac{V_n(f)}{V_0}\right) = \sum_{i=1}^n \ln\left(1 + \frac{r}{n} + f\left(X_i - \frac{r}{n}\right)\right).$$

The last expression can be simplified considerably if we first take the Taylor series expansion and, specifically, we are going to use the following fact:

$$\ln(1+z) = z - \frac{z^2}{2} + \frac{z^3}{3} - \dots + \frac{(-1)^{n+1}z^n}{n}.$$

In our case

$$z = \frac{r}{n} + f\left(X_i - \frac{r}{n}\right)$$

which yields

$$z^2 = \frac{r^2}{n^2} + f^2\left(X_i - \frac{r}{n}\right) + 2\frac{r}{n}f\left(X_i - \frac{r}{n}\right).$$

Using this expansion we see that some terms go to zero faster than others as $n \rightarrow \infty$ and we do not need terms that are smaller than $O(n^{-3/2})$ as it will soon be evident. If we take the expected value of z^2 then the only term that is of relevance to us is going to be

$$E[f^2 X_i^2] = f^2 E[X_i^2] = f^2 \left(\frac{\sigma^2}{n} + \frac{\mu^2}{n^2}\right) \approx f^2 \frac{\sigma^2}{n}.$$

Therefore,

$$G_i(f) = \frac{r}{n} + \frac{f}{n}(\mu - r) + \frac{f^2 \sigma^2}{2n} + O(n^{-3/2}).$$

Finally,

$$G(f) = \lim_{n \rightarrow \infty} \sum_{i=1}^n G_i(f) = r + f(\mu - r) - \frac{f^2 \sigma^2}{2} + \lim_{n \rightarrow \infty} O(n^{-1/2}) = r + f(\mu - r) - \frac{f^2 \sigma^2}{2}$$

This also shows why we were not interested in terms smaller than $O(n^{-3/2})$ as they would become infinitesimal and be of no consequence for the final formula as $n \rightarrow \infty$.

$G(f)$ is very easy to maximize and we find that the optimal fraction of our money to invest in a given security is maximized by:

$$f^* = \frac{\mu - r}{\sigma^2},$$

which is an expression similar to the Sharpe ratio, but not quite. It is actually the Sharpe ratio divided by the volatility of the stock. From the last formula, it emerges that if $\mu > r + \sigma^2$, then $f^* > 1$ which means that we would use leverage (i.e. borrow at the risk free rate and invest in the risky asset). On the other hand, if $\mu < r + \sigma^2$, then $f^* < 1$ and we will invest in both the risky security and a risk-free bond.

We are now sufficiently prepared to consider an example.

Example (Double Kelly).

Lets start with

$$G_p := E_p - \frac{1}{2} V_p$$

where E_p , V_p , and g_p are the portfolio expected return, variance and expected log, respectively as they were determined earlier. If we assume that the CAPM holds, we know that

$$E_p = r + (E_M - r)f, \quad V_p = \sigma_M^2 f^2$$

where E_M is the market return, and r the risk free rate and f is the portfolio weight. With a little Calculus, we find that

$$f^* = \frac{E_M - r}{\sigma_M^2}$$

which is the optimal Kelly bet with optimal growth given by

$$\begin{aligned} g^* &= r + \frac{(E_M - r)^2}{\sigma_M^2} - \frac{1}{2} \frac{(E_M - r)^2 / \sigma_M^2}{\sigma_M^4} \\ &= r + \frac{E_M - r}{2\sigma_M^2} \end{aligned}$$

If we use a double Kelly strategy, i.e., we bet $\tilde{f} = 2f$, and replace f with this $\tilde{f}$ we get

$$\tilde{G}(p) = r + \frac{2(E_M - r)^2}{\sigma_M^2} - \frac{4(E_M - r)^2}{2\sigma_M^2} = r,$$

that is the optimal growth equals the risk-free rate! Something that perhaps we would not have expected to find.

3. Optimal Growth vs. Maximum Sharpe: What's Your Objective and Why?

You will notice that the Kelly approach to investment sizing assumes that you want to maximize the expected growth rate of your wealth, that is, you are pursuing the optimal growth strategy. This is equivalent to maximizing your expected logarithmic utility and maximizing the expected value of the logarithm of your wealth. You will get a sense from this reading how the Kelly criterion fits with the log utility: The geometric compounding of returns lends itself more readily to optimization after taking the logarithm.

The optimal growth strategy is also quite distinct from the strategy that leads to the optimal Sharpe portfolio, and yet the Sharpe ratio is probably the most popular metric for comparing strategies. For this reason, it is worth discussing the optimal Sharpe portfolio and the mean-variance optimization framework that it rests on.

The Sharpe ratio seems like a natural choice for evaluating portfolios because it acknowledges the inherent trade-off for investors between risk and return: Since we all want high (excess) returns and low risk, the optimal portfolio appears to be the one that maximizes the ratio of excess risk to volatility. In this framework, higher volatility in returns can be offset by higher excess returns. The problem of finding the optimal Sharpe portfolio can in turn be reduced to the well-known mean-variance optimization problem introduced by Markowitz (and discussed in Module 1). Indeed, the Sharpe optimal portfolio sits on the point of tangency of between the capital allocation line (CAL) and the efficient frontier (also discussed in Module 1). However, when the optimal portfolio is a function of expected utility that only accounts for the mean and the variance of returns, this utility function is considered quadratic.

One question worth asking is whether the Kelly criterion might be the last word. The answer is that, generally, speaking the Kelly criterion has its own disadvantages. The main disadvantage of the Kelly is that its suggested wagers may be very large and hence, the Kelly criterion can be very risky in the short term.

To this purpose, we have empirical research that shows that Kelly criterion has superior long-run growth properties, but can still have a poor return outcome. Ziemba and Hausch in a 1996 paper have shown that on $n = 700$ wagers, with a 14% advantage, a full Kelly can turn \$10,000 into at least \$100,000, 16.6% of the time, but there is also a chance to get only \$18. On the other hand, a half Kelly strategy, showed that \$1,000 can turn into \$145, but the probability to exceed \$100,000 is only 0.1%. Furthermore, it can take a long time for any strategy including Kelly, to dominate an essentially different strategy.

Another criticism is that the theory and practical application of the Kelly criterion is straightforward when the underlying probability distributions are fairly accurately known. However, in investment applications this is usually not the case. Realized future equity returns may be very different from what one would expect using estimates based on historical returns. Consequently practitioners who wish to protect capital above all, sharply reduce risk as their drawdown increases.

For long term "compounders", the good properties dominate the bad properties of the Kelly criterion. But the bad properties may dampen the enthusiasm of less experienced investors about making investment decisions based on the Kelly criterion. Nevertheless, the Kelly and fractional Kelly strategies are very useful if applied carefully with good data input and proper financial engineering risk control.

Arguably, one of the most interesting criticisms is that according to which individual's investment constraints may override the desire for an optimal growth rate. These criticisms could be countered by recognizing that the main concept of expected utility maximization is actually in place since the Kelly criterion would fall under expected utility maximization when the utility function is a logarithmic utility function as we explained earlier.

Students who are interested in this type of topics should consult the reading by MacLean, Thorp and Ziemba that appears in the reference section.

4. Logarithmic vs. Quadratic Utility

We would like to review the main implications of adopting a logarithmic utility function vs. a quadratic one to better appreciate the relative strengths of the logarithmic utility function and the optimal growth strategy that results from the Kelly Criterion. As you recall from Module 1 in this course, there are many utility functions. Log and quadratic utility functions are perhaps so popular among these thanks to their tractability: They are easy to work with in mathematical formulas. But recall from module 1 the main drawback associated with quadratic utility: it suggests that an investor's absolute and relative risk aversions increase as wealth increases. It is unrealistic or at least counterintuitive that investors would feel less and less comfortable investing in risky assets as their wealth increases. If anything, one might expect an investor to have more risk appetite as their wealth increases, on an absolute or relative basis: If I am willing to invest ten Euros in a risky asset when I only have 100 Euros (10 Euros on an absolute basis; 10% on a relative basis), then I am most likely willing to invest at least 10 Euros in the same risky asset when I have 1000 Euros (same 10 Euros on an absolute basis, but now only 1% on a relative basis). However, according to the quadratic utility function, you would not even be willing to invest the 10 Euros given your 1,000 Euro wealth—even though you were willing to do so before. And yet this unreasonable preference regarding risk aversion is the only one that makes mean-variance optimization consistent with maximization of quadratic utility.

The logarithmic utility function, on the other hand, implies decreasing absolute risk aversion. So as my wealth increases from 100 to 1000 Euros, I am more willing to invest 10 Euros in a risky security. This seems more plausible. The logarithmic utility function also implies constant relative risk aversion, so if I am willing to invest 10 Euros in a risky security when I have 100 Euro wealth (10%), then I would be willing to invest 100 Euro in a risky security when I have 1000 Euro wealth (10%). This constant relative risk aversion is not necessarily the only rational approach to utility and risk management; however, it does at least seem plausible, while decreasing relative risk aversion does not appear generally realistic.

5. Kelly criterion for multivariate portfolios

This is a necessary extension of the topic as we cannot really go very far investing one asset at a time, except in very particular and not very realistic cases.

In presenting the relevant result, we will follow a paper by V. Nekrasov (2013) The setup is a bit more complicated than before as now we need to decide on the fraction of wealth to invest for n risky assets and not just 1.

We assume that the initial wealth is $V_0 = \$1$ which does not cause any loss of generality. We call f_k $k = 1, 2, \dots, n$ the fraction of wealth invested on each of the n assets with the understanding that any residual capital $1 - \sum_{k=1}^n f_k$ is invested in a risk-free bond. Therefore, after the first trade our wealth is given by

$$V_1(f_1, \dots, f_n) = \sum_{k=1}^n f_k(1 + r_k) + (1 + r)(1 - \sum_{k=1}^n f_k) = (1 + r) + \sum_{k=1}^n f_k(r_k - r).$$

with r_k , $k = 1, 2, \dots, n$ being the expected returns on each of the n assets and r denoting the risk-free rate. In this case, the goal is to find

$$\max_{f_1, \dots, f_n} E \left[\ln \left((1 + r) + \sum_{k=1}^n f_k(r_k - r) \right) \right].$$

We know already that maximizing the expected terminal utility is equivalent to maximize the function above on each trading instance. If we expand the function above in a Taylor series about $\mathbf{f}_0 = (\mathbf{0}, \mathbf{0}, \dots, \mathbf{0})$, then the problem becomes one to maximize the following

$$E \left[\ln(1 + r) + \sum_{k=1}^n \frac{f_k(r_k - r)}{1 + r} - \frac{1}{2} \sum_{k=1}^n \sum_{h=1}^n f_k f_h \frac{(r_k - r)(r_h - r)}{(1 + r)^2} \right]$$

which can be written in a simpler manner in matrix notation as

$$\ln(1 + r) + \frac{1}{1 + r} (\mu - \iota r)^T \mathbf{f} - \frac{1}{2} \frac{1}{(1 + r)^2} \mathbf{f}^T \Sigma \mathbf{f}$$

with $\mathbf{f}^T = (f_1, f_2, \dots, f_n)$, $\mu^T = (\mu_1, \mu_2, \dots, \mu_n)$, are the means of the expected returns for $r_1, r_2, \dots, r_n$,

$\iota = (1, 1, \dots, 1)$,

and Σ is the variance covariance matrix for the returns of the n assets. One should be able to recognize that this is a QP optimization problem. If we know μ

and Σ , in the absence of constraints, this problem can be solved analytically and it yields:

$$\mathbf{f}^* = (1 + r) \Sigma^{-1} (\mu - \iota r).$$

When constraints are present, it is no longer possible to solve the problem in such a simple manner and numerical routines are necessary.

An interesting result is found when we make the strong assumptions that the assets are not correlated, i.e. the variance-covariance matrix is diagonal $\Sigma = \text{diag}(\sigma_1^2, \sigma_2^2, \dots, \sigma_n^2)$. In fact, in this case, we get

$$f_k^* = (1 + r) \frac{\mu_k - r}{\sigma_k^2} \quad k = 1, 2, \dots, n,$$

which is very similar to the formula found in the univariate case, except that in this case it covers n fractions instead of just one and it contains a factor $(1 + r)$ which for the typical value of the risk free rate is not very different from 1. The difference is due to the fact that earlier we derived the formula under the assumption of a log-normal diffusion for stock prices, Here we made no assumptions about the distribution. So the result that we just found is model-free (non-parametric). In general, we will need to estimate μ and Σ from the returns of our assets during a relevant period.

6. Sharpe is Near-Sighted

Other issues with the Sharpe ratio and the optimal Sharpe portfolio is myopia (nearsightedness): The mean-variance framework is always only sizing the current investment. It does not optimize the investment size for anything beyond the current set of investment opportunities and parameters. As you read about the Kelly criterion, on the other hand, you will see that it takes into account the reinvestment of returns over a multi-period horizon. This compounding of returns is a geometric calculation, as opposed to the arithmetic sum of returns that Markowitz and Sharpe implicitly take for granted.

Note: Be sure to review all the code referenced in the required reading article, including the Karan Chawla's Github repo devoted to the Kelly criterion.

7. Conclusions

In this lesson, we discussed the optimal growth strategy and how the Kelly criterion helps to achieve it. We also discussed the mean-variance optimization strategy and how the Sharpe ratio naturally follows from it. We also reviewed the advantages and disadvantages of the quadratic and logarithmic utility functions. We found the implications of the logarithmic utility function to be more realistic, which appears to be one more reason to prefer the optimal growth strategy to the optimal Sharpe portfolio.

In the next lesson, we look at some recent scholarship on the subject of the Kelly criterion, providing more theoretical background as well as empirical findings and innovations to control for risk.

References

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- Nekrasov, Vasily (2013). "Kelly Criterion for Multivariate Portfolios: A Model-Free Approach". SSRN 2259133.

